Stabilization baseline and release gates

SolariNet and RackWire | Baseline 20 October 2026 · candidate evidence updated 21 October 2026 | operational snapshot, not a release declaration

The existing August 9 stabilization workflow remains the controlling plan. This snapshot records what was checked on October 20, the first integration changes, and the gates that still prevent a trustworthy platform release.

Release state, rechecked 2026-10-21: No new SolariNet or RackWire version is ready to tag. GitHub and Forgejo main still match at `5e8d688`; `integrate/stability-20261020` is 18 commits ahead and has the separate `stabilize/phase-0-containment` branch as an ancestor. The integration worktree has 176 root tracked paths changed and 123 untracked paths, plus a dirty RackWire submodule. The containment worktree is 31 commits behind and 2 ahead of main; do not use it as the release candidate. Root and RackWire have no tags. The candidate gitlink advances RackWire to its Forgejo main at `5f29f59`. Scanner-ingest has a local Git root, but no GitHub/Forgejo remote. Independent review, production security and migration gates, and release ownership remain open.

1. Observed state

Surface	Evidence from this session	Implication
Benzene RabbitMQ	**OBSERVED · 2026-10-21** The host `rabbitmq-server` systemd unit is inactive because the broker is container-managed: rootful Podman runs `rabbitmq:3-management` with a persistent data volume and `unless-stopped` policy. A read-only recheck confirmed RabbitMQ 3.13.7 fully booted without alarms, with AMQP/management/Prometheus/clustering listeners and four running application queues (one consumer each; zero ready or unacknowledged messages); six client connections were observed. In vhost `solari`, all four application queues are durable classic queues, non-auto-delete, empty, and each has one consumer. Khepri is disabled. The current container tag is unpinned. RabbitMQ 3.13 community support ended 2024-09-30; the 4.3 upgrade guide requires 4.2 as an intermediate. A fresh isolated SoR-sync run passes 37/37 runnable tests against MariaDB 11.8.6 and RabbitMQ 4.3.6; the suite's separately gated disk-upgrade test is skipped there. The dedicated synthetic volume rehearsal passes 3.13.7 → 4.2.9 → 4.3.7 → 4.4.0, including a durable message and pre-4.4 snapshot restore. Separately, the verified 287744-byte Benzene backup matches its root-only Xenon copy; a network-disabled clone of that actual data booted on 3.13.7 and preserved all four durable empty queues through 4.2.9 and 4.3.7 without alarms. Because the actual snapshot had no queued messages, this proves queue-definition continuity only. The official upgrade guide currently documents 3.13 → 4.2 → 4.3; RabbitMQ 4.4.0 release notes warn that new `.sqs` files cannot be downgraded to 4.3 or earlier. Defer live cutover pending independent review and pinned startup; defer 4.4 until its supported rollback path is established. Commercial-support entitlement was not checked.	Broker health is good; the live 3.13 tag floats and is outside community support unless commercial support applies. The actual-volume clone upgrade through 4.3.7 and backup restore are verified. Before live cutover, pin startup, complete independent review, and verify host-reboot recovery and a real client message flow. Defer 4.4 until its supported rollback path is established. No production service changes were made.

Surface	Evidence from this session	Implication
Repositories	**RECHECKED · 2026-10-21** SolariNet GitHub and Forgejo main both resolve to `5e8d688`. The current local integration branch is 18 commits ahead of both remotes. Its RackWire gitlink and submodule HEAD are `5f29f59`, matching RackWire Forgejo `origin/main`; the submodule working tree has pending documentation and entrypoint edits. Published SolariNet main pins `fccf5ff`. Neither root nor RackWire has release tags.	Keep the integration in a review branch until its tests and cross-app behavior pass.
Live services	**OBSERVED** Current read-only checks show xenon BIND, server, client, monitor, SoR emitter, DNS applier, Pi-hole applier, and discovery reconciler active; its `solarinet-dashboard` systemd unit is inactive. The SoR primary is cesium (`10.1.0.200:3306`) with MariaDB active. RabbitMQ is a root-owned Podman container on benzene, not cesium. The deployed DNS applier still differs from this candidate; the release helper/config are absent. Jason identifies the Steel inventory label on the ZimaBlade, on port 12 of `laboratory`; it booted without useful services and remains unreachable by SSH. DNS no longer advertises .11 as a nameserver. The separate Khadas VIM4 is `chlorine` (`10.7.0.10`): BIND 9.18.39 is active as an authoritative-only secondary for 109 zones: `akoria.net`, 17 generated reverse zones, 90 bare-name zones, and `rpz.akoria`. Xenon's `named-checkconf -z` passes and chlorine's BIND config passes. Refresh verifies matching SOA serials for all 109 zones plus Radium's forward zone. Canonical `named-compilezone` hashes match for all 109 primary/secondary zones. Direct queries confirm authoritative `dashboard` and RPZ responses, chlorine in the `akoria.net` NS set, and refused recursion. BIND listens only on `10.7.0.10:53`; `step-ca` is active. Xenon watches the primary fragments and RPZ data with a five-minute retry. Bare-zone/RPZ replication is now active; the legacy SoR applier's updated generator has not yet been exercised by a live event. The operator identifies three physical Zima hosts: Steel-labeled ZimaBlade on laboratory switch port 12, ZimaBoard, and ZimaBoard 2. The ZimaBlade booted without useful services and is unreachable by SSH; imaging requires console access, and DNS does not advertise .11. The ZimaBoard and ZimaBoard 2 have no confirmed hostname/IP mapping yet. Chlorine already provides the backup authoritative BIND service; Steel's tertiary role awaits console redeployment.	Keep the ZimaBlade .11 handoff deferred until console redeployment. Map the ZimaBoard and ZimaBoard 2 to records, exercise the live SoR generator, validate client resolver failover, and complete cross-lab review before broad rollout.

Surface	Evidence from this session	Implication
RabbitMQ on benzene	**READ-ONLY RUNTIME CHECK · 2026-10-21** Host systemd has no RabbitMQ unit; Podman runs `rabbitmq:3-management` as a container. RabbitMQ ping and running checks succeeded, local resource alarms are clear, and AMQP/management TCP ports 5672/15672 accept connections. An earlier same-day check found four queues with zero ready/unacknowledged messages, one consumer each, and six broker connections. At 12:34 UTC, a fresh read-only check again found RabbitMQ ping and running checks successful, no local alarms, all four expected ports (5672, 15672, 15692, 25672) reachable, and the container at 0.17% CPU with zero restarts since 2026-08-21. The first queue query targeted the CLI default vhost `/`; querying the configured `solari` vhost explicitly returned all four queues with zero ready/unacknowledged messages and one running consumer each. The successful explicit query confirms the configured application vhost is healthy; the direct reason the initial default-vhost query timed out remains unverified. Xenon's SoR emitter, DNS applier, Pi-hole applier, and reconciler are active, and recent reconciler log entries report normal completion.	Broker runtime and current consumer attachment look healthy. Empty queues and active consumers do not prove a fresh event traversed every target; exercise the deployed end-to-end delivery path before release.
Database schemas	**OBSERVED** Read-only full schema comparison on 2026-10-21: Xenon `solarinet` has 29 tables versus the 33-table candidate, lacks `ctlRequest`, `decommissionChallenge`, `discoveryScanJob`, and `reportReceipt`, seven enrollment/node fields, the token-hash unique index, and the singleton check; one old approved enrollment has no matching node and cannot be auto-backfilled. It has one extra maintenance-window index pending query-plan review. Cesium `sor` has 41 tables versus 45, lacks four migration-021 tables, two generated IP columns, two unique indexes and two IP-address checks, and still has `ON DELETE SET NULL` on `fk_ip_entity`. Its two CDC tables and 24 triggers are present. Four SoR view definitions were queried directly and match candidate logic. Neither database has a migration ledger; no production SQL was applied.	Do not adopt or upgrade until historical migration provenance, the orphaned enrollment disposition, and a Xenon monitoring backup/restore are resolved. Candidate enrollment and leaf-certificate identity persistence are not production-ready against Xenon's current schema.
Dashboard static deploy	**OBSERVED · 2026-10-21** The served `api.jsx` alone differed from the repository and lacked the required CSRF header. The repository file was atomically installed on xenon, its rollback was rehearsed, and the served SHA-256 now matches `ce0861ef...`. Candidate Apache config sets HSTS, referrer, framing, MIME-sniffing, and root-shell CSP headers; isolated syntax and HTTPS checks pass. Read-only requests to the live root and RackWire both return 200 only with certificate verification disabled. The live dashboard certificate is self-signed and distinct from Akoria Root CA. Current live responses include only `nosniff` and `SAMEORIGIN`; candidate HSTS, referrer, and CSP headers are absent.	Do not install the candidate vhost until its TLS certificate is issued through the approved trust path and browser behavior is tested. The candidate config and dashboard release remain undeployed.

Surface	Evidence from this session	Implication
PHP/C privilege boundary	**OBSERVED** The candidate has a separate `www-solari` PHP-FPM identity, `SO_PEERCRED` verb ACL, server-owned privileged request queue, least-privilege dashboard DB grants that exclude `ctlRequest`, and a disposable end-to-end boundary verifier. The production README now selects the hardened configuration and supplies a parser-checked server config template. The live Xenon PHP and server services have not been cut over and remain on the legacy same-uid configuration.	Complete the versioned release-tree installer, independent review of the latest deployment/config changes, and rollback rehearsal; then verify the live uid, socket, DB grants, and privileged queue behavior. Until that cutover passes, F-04 remains a production blocker.
RackWire	**VERIFIED LOCALLY · 2026-10-21** The standalone discovery suite passes 45/45, the parent API contract passes 58/58, the integrated theme/selection and barcode-CSRF bridge passes 51/51, dashboard JSX parsing passes 16/16, and vendored runtime checksums pass 4/4. The standalone headless browser smoke covers offline `file://` operation, accessible dialogs and focus handling, local gear placement/reload, Map/Data navigation, and narrow-toolbar layout. The authenticated scanner-to-RackWire browser E2E passes against a disposable SoR and cached Chromium: scanner import, model profile save, embedded barcode lookup with CSRF, asset tag/serial placement, plan persistence, and scan admission. Its controlled API fixture now exercises pending admission, live progress, disabled repeat submission, navigation away/back persistence, and completion. A separate full process/API/SoR acceptance run proves the native worker scanning a real loopback listener, authenticated read/adoption of that resulting candidate, C bridge asset/service creation, and SoR host/address mirror; all 13 HTTP/persistence checks passed against fresh isolated databases migrated through 34 monitoring and 8 SoR migrations. RackWire manuals describe network, power, and interconnect planning, optional rack layouts, and the distinction between monitoring discovery and physical inventory.	A real lab-subnet scan, browser polling against the actual worker API, deployed UI/telemetry, production SoR commit/rollback, cross-lab review, and hosted CI remain open; no deployment or release was performed.
DNS and certificates	**OBSERVED** Xenon's read-only `named-checkconf -z` passes; the authoritative `akoria.net` SOA is serial `2026100301`, and `dashboard.akoria.net` resolves to `10.0.0.20`. `rackwire.akoria.net` is NXDOMAIN locally and publicly, while the canonical dashboard path works. Public `akoria.net` certificate expired July 19; certbot fails ACME challenges.	Maintain current DNS while preparing a staged cutover. Decide whether the separate RackWire hostname is a published surface. Repair public ACME reachability or use DNS-01 before calling the public site healthy.

2. Verification completed

- **OBSERVED** The candidate builds with server, I/O, SQLite, and JSON enabled; all 33 C test suites pass, including the loopback transport, enrollment installer, CSR signer, and control ACL suites (rechecked 2026-10-20).
- **RECHECKED LOCALLY · 2026-10-21** A fresh disposable MariaDB 11.8.6 run passes DB-backed CTest 37/37, with the replacement-certificate gossip test stable across ten consecutive runs after replacing its fixed-delay assertion with bounded polling. The tool-enabled scan-worker process and authenticated control-queue tests pass; the loopback browser flow completes RackWire handoff, live scan progress, navigation recovery, and candidate ingest. A two-server/two-worker process test runs enrichment during active lease renewal, then fails over and verifies stale-worker fencing, orphan recovery, and a successor scan; this check is wired into CI. Fresh-SoR scanner preview/apply, HTTP authorization, rollback/idempotency, and scanner-to-RackWire barcode/profile/plan browser acceptance pass, as does HTTPS enrollment and isolated BIND AXFR/NOTIFY refresh. Monitoring and SoR migration histories (34 and 8) match their generated snapshots and restored object sets. All fixtures were removed. ShellCheck 0.11.0 passes all 35 repository shell scripts with sourced-file analysis and warning severity; hosted CI, independent cross-lab review, and deployed behavior remain open.
- **VERIFIED LOCALLY · 2026-10-21** A fresh native server build passes CTest 37/37. Against disposable MariaDB 11.8.6, all 34 monitoring and 8 SoR migrations apply and pass clean recheck; generated snapshots match byte-for-byte, restored snapshots match the migrated table/view sets, and SoR triggers match. The complete dashboard CI job passes locally, including PHP syntax, authorization and route audits, dashboard/UI suites, two-session freshness browser acceptance, scanner import plus RackWire iframe save/readback, release packaging/rollback, ledgered clean installs, schema export parity, deployment script syntax, and the enrollment identity contract. The authenticated scanner import preview handles 5,000 active assets under PHP's 128 MiB memory limit without writing inventory. MCP (86), authbroker (6), Opie (5 pass/1 skip), backup (10), DNS bare (4), Steel DNS (1), and notifyd (3) suites pass; SoR sync passes 37 runnable tests against isolated RabbitMQ 3.13.7 and 4.3.6, and the separate RabbitMQ persistent-volume upgrade/restore rehearsal passes. Disposable containers were removed; live Xenon and SoR databases were not used.
- **VERIFIED LOCALLY · 2026-10-21** Added a browser-level scanner-ingest changeover test against an isolated MariaDB 11.8.6 SoR and authenticated PHP dashboard. It verifies dashboard bootstrap, duplicate-name location choice by SoR row ID, that preview makes no inventory write, fingerprint-bound apply, persisted location/asset readback, and immediate appearance in the refreshed Inventory view. This run exposed and fixed the missing combined `GET /api/inventory` read route and the dashboard's camelCase-to-snake_case normalization gap. The prior browser run verified embedded RackWire autosave with CSRF and SoR readback. The expanded acceptance now exercises imported model profiling, RackWire library merge, and barcode scan placement with the asset tag and serial. Its PHP HTTP route equivalent passes against a disposable local MariaDB, including viewer and missing-CSRF rejection. The expanded browser test passed locally with the cached Playwright Chromium; its four-phase progress and final candidate state were observed in the UI. No production SoR or network device was used.
- **VERIFIED LOCALLY · 2026-10-21** Packaged the real candidate server, scan worker, dashboard API/SPA, and pinned RackWire checkout into a 179-file preflight archive. The temporary-root installer verified its hashes and activated the release, including the RackWire entrypoint and worker binary. Its metadata correctly marks the source dirty; this is package/install evidence only, not a production release or deployment.
- **RECHECKED LOCALLY · 2026-10-21** RackWire's standalone discovery backend passes 45/45, its four vendored runtime checksums verify, and the parent API/UI contracts pass 58/58 and 51/51. Dashboard release-package/install/rollback tests pass 8/8 and migration/schema tools pass 17/17.
- **OBSERVED** The full server was built with pinned mbedTLS 3.6.2 and NNG 1.8.0 under ASAN/UBSAN. All 32 CTest entries passed against a disposable migrated MariaDB, including the live DB integration; that DB suite also passed all 12 cases against a separate MariaDB 11 container. The ASAN/UBSAN loopback transport test passed separately. ShellCheck passed all 30 shell scripts, and the pinned-dependency bootstrap scripts pass Bash syntax checks.

- **VERIFIED LOCALLY** A new optional Clang/libFuzzer target instruments the shared frame, TLV, and message parser library. Its deterministic 20,000-run smoke passed with the protocol dictionary, reaching 469 coverage counters and 781 feature tuples without a sanitizer finding (105 corpus additions; 41 MiB peak RSS). The full opt-in Clang build and its nine CTest entries pass with `-Werror`. A Clang-specific diagnostic suppression is scoped to the Linux PAL translation unit for glibc's ISO C23 string macro expansion under C99. CI now builds and tests all nine targets with warnings treated as errors; remote CI has not run for this candidate.
- **RECHECKED LOCALLY · 2026-10-21** The current candidate's optional native I/O loopback passed separately, the Clang `-Werror` fuzz/core build passed all nine CTest entries, and ShellCheck 0.11.0 passed all repository shell scripts with sourced-file analysis enabled.
- **VERIFIED LOCALLY** SCP frame parsing and building now enforce the same envelope: valid magic/version, zero reserved byte, supported flags (compression is rejected because no decoder exists), complete TLVs, and an exact declared element count. Unknown TLV types remain accepted. Focused rejection tests, the Clang `-Werror` fuzz suite, and the full server-enabled CTest suite pass; cross-lab review and remote CI remain pending.
- **CANDIDATE AND LIVE BEHAVIOR CHECKED · 2026-10-21** The candidate dashboard HTTPS vhost emits six-month HSTS, `no-referrer`, `SAMEORIGIN`, and `nosniff`, and scopes same-origin CSP to the main SPA while leaving RackWire's inline document outside it. Isolated Apache syntax and HTTPS checks pass. Live HTTPS requests to both paths return 200 with `curl -k`; certificate verification fails because the dashboard leaf is self-signed, not issued by Akoria Root CA. Live responses contain `nosniff` and `SAMEORIGIN` only; HSTS, referrer, and CSP are absent. No server configuration was changed.
- **OBSERVED** PHP route/authorization tests pass; the full dashboard PHP and JS suites pass after the RackWire token-contract correction.
- **VERIFIED LOCALLY** The dashboard certificate screen now requests `https://chlorine.akoria.net` with TLS chain and hostname verification. The internal root is already installed in Xenon's system trust store; a live PHP request reports step-ca healthy. A hermetic HTTPS test rejects an untrusted certificate and a plaintext HTTP endpoint, and accepts a trusted endpoint using an explicit CA bundle (3/3 pass). This code change is not deployed.
- **OBSERVED** A disposable MariaDB 11 schema accepted all 34 monitoring migrations, and the live DB integration suite passed, including request-queue and durable report-replay behavior. A fresh 2026-10-21 clean install and ledger check also passed; schema snapshots matched both migrated and restored object sets.
- **OBSERVED** A service-worker harness proved that offline API requests reject instead of reading an old authenticated cache, and activation removes the prior v6 API cache.
- **OBSERVED** The F-11 control bridge candidate now requires a complete newline-terminated request within a 16 KiB cap and two-second deadline. The PHP and C clients loop on short writes and reject incomplete replies. A socket-pair test exercises fragmented, 8 KiB, and truncated requests; C suites and PHP parser checks pass.
- **VERIFIED LOCALLY · 2026-10-21** The complete SoR-sync suite passes 37/37 runnable tests against isolated MariaDB 11.8.6 and RabbitMQ 4.3.6 (one disk-upgrade test is gated separately), including mandatory routing before checkpointing, heartbeats during slow applies, redelivery, configured-secondary retry/status, fail-closed symlinked state, and daemon recovery after RabbitMQ application and container restarts. The dedicated volume-upgrade test passes 3.13.7 → 4.2.9 → 4.3.7 → 4.4.0 and pre-4.4 snapshot restore. The DNS chain test creates a CDC-triggered entity/IP outbox event, publishes it through RabbitMQ, renders with the real SoR DNS generator, invokes the real release CLI through `sudo -n` with a root-owned temporary config/tree, and checks its served answer and target state before acknowledgment. The private primary BIND runs as `bind`; the secondary refresh is a controlled test helper, not live chlorine. An isolated consumer disconnect and a separate-process SIGKILL after apply but before acknowledgment prove requeue and retry. Seventeen isolated BIND activation/cutover tests pass, including NS-only verification and activation under umask `077`. Host reboot, automatic systemd consumer restart, production backup/restore, production cutover, and real secondary-chain acceptance remain open.
- **VERIFIED LOCALLY · 2026-10-21** The DNS candidate now requires explicitly named IPv4 secondaries, renders AXFR/NOTIFY ACLs only for their configured addresses, refreshes each secondary after Xenon's verified primary release, and waits for SOA parity before acknowledging. The legacy direct Steel copy target is retired in durable target state; the offline ZimaBlade is not a default. Unit coverage verifies target validation, secondary failure retention, retries, and acknowledgment only after secondary proof. The bare-zone generator passes 4/4 tests with chlorine-only defaults; the Khadas refresh/updater suite passes 6/6. These are candidate and isolated tests: the live legacy writer remains active, and no production DNS state changed.

- **OBSERVED** Dashboard PHP/JS checks pass, including 57 RackWire API assertions, 33 token-bridge checks, lifecycle checks, panel UI checks, and 59 page/firmware parity checks. The status-panel firmware host suite passes 107 assertions; the daemon listener passes 16 TLS cases. The provisioning smoke passed ten consecutive local runs, including dropped-ack and desync recovery.
- **VERIFIED LOCALLY · 2026-10-21** The F-12 migration candidate installs 33 monitoring and 8 SoR migrations in disposable MariaDB databases and passes clean rechecks. A fresh local MariaDB 11.8.6 run independently passed both migration ledgers, confirmed the expected row counts, reproduced both generated snapshots byte-for-byte, and matched the migrated and restored table/view sets and SoR triggers. Both complete schema snapshots are generated deterministically and restore into clean databases. A fresh MariaDB CI-style run found account-dependent DEFINER clauses in view/trigger dumps; `export_schema.py` omits those identities while retaining SQL SECURITY, with regression coverage. The fresh HTTPS enrollment and C server DB integration also pass against disposable MariaDB. The runner refuses an untracked populated database and an interrupted ledger row; `--init-db` fails explicitly. The read-only `tools/audit_schema.py` compares table, view, trigger, routine, and event definitions with the candidate snapshot, excludes the migration ledger, and emits a unified diff; its hermetic tests pass. A disposable MariaDB run applied all 33 monitoring and 8 SoR migrations, reproduced both snapshots byte-for-byte, matched both clean schema audits, detected injected drift, and rejected a wrong-collation database before DDL. Read-only metadata checks confirmed Xenon and Cesium use the canonical database default; production adoption remains open. See the standalone migration runbook and PDF.
- **OBSERVED · RECHECKED 2026-10-21** Cesium's `solari-backup@cesium.timer` is enabled and active, with a 21 October SoR archive present. The latest archive passed gzip integrity and a fresh restore into disposable local MariaDB; all 41 base tables pass `mariadb-check`, with four views and 24 triggers. Xenon has no enabled SolariNet backup timer/config/runner or matching database dump in checked roots. Its existing `/var/backups/solarinet` is root:root mode 0750 and holds a dashboard rollback file, not a DB archive. The monitoring database is 658 MiB; Xenon's local NVMe has 372 GiB free, and MariaDB socket auth as OS user `mysql` succeeds. Read-only NAS-X inspection confirmed a private Jason home share; a candidate off-host mount/config is prepared, but its dedicated directory has not been created and effective CIFS permissions, backup, and restore remain unverified. The backup runner requires a root-owned `0700` target and publishes root-owned `0600` artifacts; Xenon's example config also requires a mounted filesystem. All ten hermetic tests passed, covering backup atomicity, privacy, retention, mount enforcement, dump failure, AMQP notification publication, missing-Pika fallback, and credential-safe notification errors.
- **OBSERVED** The F-14 auth candidate gates local password login at the API and requires a private rate-limit directory; negative tests cover lockout, stale OIDC state, missing token claims, session expiry, and local role downgrade. The complete dashboard PHP suite passes. Live SSO callback, break-glass operation, and browser session tests remain open.
- **VERIFIED LOCALLY** Notifyd now validates its JSON envelope before dispatch: text and provenance types, supported severity, non-empty routing channel and recipients, and bounded numeric timestamp. Three parser tests cover valid messages, missing summaries, malformed fields, NaN/infinity, and datetime overflow; the suite is wired into CI. Sender-specific transport checks remain open.
- **VERIFIED LOCALLY** ShellCheck with source-following enabled passes all 29 `.sh`/`.bash` files. The gate is in CI; it fixes one unchecked directory change, removes one unused variable, and documents two globals consumed by fleet script callers.
- **VERIFIED LOCALLY** Authbroker's MQTT approval test defaults to anonymous loopback and contains no broker credential default. Six unit tests pass; approve and deny round trips both pass against a disposable Mosquitto container. CI now repeats both decisions against a listener bound only to loopback; its authbroker dependencies are pinned in `deploy/authbroker/requirements.txt`, and no live broker was contacted.
- **VERIFIED LOCALLY · 2026-10-21** With the integration's declared dependencies installed in an isolated temporary virtual environment, the MCP package's complete pytest suite passes (86 tests), its 10 fixture evaluations pass, and `python -m solarinet_mcp --check` lists all 26 tools without connecting to configured services. Authbroker's six unit tests also pass with its pinned requirements installed in a separate temporary environment; the live broker round-trip was not repeated in this pass.

- **OBSERVED** Enrollment approval requires the CSR CN to match the approved `role.fqdn` and exactly one DNS SAN to match the approved host; the issued leaf preserves that SAN. Ingest binds each frame's `sourceNodeId` and exact certificate CN to the approved node row and fails closed on lookup errors. Unit, installer, and HTTPS fixture tests cover identity rejection, forced renewal recovery, request validation, and acknowledgement. On 2026-10-20 the full HTTPS → PHP → `solariCtl` → TLS-enabled C server → migrated disposable MariaDB → temporary CA flow passed locally: token submission, approval transaction, signed certificate pickup and chain validation, fingerprint/serial persistence, acknowledgement, and token/CSR cleanup. It is wired into full-server CI; remote CI is pending. Migration 024 stores leaf fingerprint and serial atomically with approval, but live leaf fingerprint enforcement remains open because the deployed NNG 1.x API lacks peer DER access. The transport wrapper now explicitly requires NNG's `TLS_VERIFIED` result to be true before accepting the peer CN; tests pass for trusted peers, reject a revoked leaf, and reject a plaintext pipe as having no verified peer identity. A loopback NNG/mbedTLS integration test now revokes a historic monitor leaf and confirms that a replacement with the same CN but a different serial remains accepted. Candidate CRL generation is implemented; production freshness enforcement and fleet distribution remain open. A scratch MariaDB 11 migration/test run now proves retired nodes fail the identity authorization lookup and a successful enrollment upsert reactivates the node. The pinned dependency bootstrap also passed on GCC 15 after its script narrowly exempted mbedTLS 3.6.2's upstream fixed-size label initializer warning from error promotion. Stale-certificate handling, authenticated monitor membership, and cross-lab security review remain release gates.
- **VERIFIED LOCALLY · 2026-10-21** Migration 025 adds durable single-use decommission challenges. Migration 026 adds a per-node generation; confirmation consumes the challenge and publishes that generation in `TLV_LIFE_GENERATION`. The server handles success acknowledgements separately from config convergence and retires only after the TLS-authenticated node returns the matching generation. Migration 027 adds persisted delivery attempts and bounded exponential retry scheduling. Migration 029 adds the operator-attributed revocation ledger, 030 records leaf issuer identity, 031 inventories issued certificates, 032 generates CRLs, and candidate migration 033 redacts legacy queued confirmation tokens and fails pending or claimed decommission requests for safe transition. Decommission now queues a token-free intent; the C queue consumer executes both challenge steps in memory and stores only the final response. PING advertises the required bridge capability and the dashboard refuses the operation against older bridges. The C control unit suite passes with an end-to-end queue consumer test, token-queue rejection, and capability checks. A new HTTP integration test drives the operator route through the real C queue process and a fresh migrated scratch MariaDB; it verified one token-free request and result and a consumed challenge, without an attached PUB transport. PHP syntax and authorization checks pass, as do the migration runner's eight hermetic tests. Migration 033 also passed a disposable socket-only MariaDB fixture covering pending and claimed requests, token-bearing historical results and errors, and preservation of unrelated queue rows. The earlier full schema run covered 33 migrations; migration 033 has not been applied to Xenon. The client still has no `SCP_MSG_DECOMMISSION` receiver, so remote teardown cannot complete. Cross-lab review, client receiver, deployed migration, production rollout, and physical wipe remain unverified.
- **VERIFIED LOCALLY · 2026-10-21** The F-13 candidate marks retained core data stale with its age and pauses writes after a failed poll. Optional integrations expose fresh/stale/unknown state and last success/attempt, retain their last successful payload during an outage, and show a partial-data banner. Adapter tests cover optional-source outage, retained values, timestamps, and recovery. The service worker bypasses API caching, and the retired SSE route returns 204; the dashboard continues polling every 10 seconds. A Chromium acceptance with two independent authenticated PHP sessions verifies stale warning, blocked writes before transport, isolation of the unaffected session, and recovery; it is wired into the existing dashboard CI job. Production outage behavior and broader authenticated auth/session flows remain unverified.
- **OBSERVED** F-10 candidates persist full client config before epoch activation and commit report receipts with current/history rows. Fault tests cover failed saves; live DB tests prove replay after reconnect/window eviction, sender sequence reuse, failed-write retry, and single monitor history writes. A candidate report ACK path now dispatches CLIENT_REPORT, MONITOR_REPORT, and SURVEY_RESP on the authenticated control REQ/REP link, and removes spooled reports only after matching sequence/session ACK validation. The client and monitor loopback ACK tests pass. Real server DB commit-failure/ACK-loss drills, failover, compatibility, coordinated fleet rollout, and Windows durability remain open.

- **VERIFIED LOCALLY · 2026-10-21** Jason added `~/Code/scanner-ingest` to project scope. Its Python/SQLite intake app includes an insert-only SoR preview/apply exporter with provenance. The full **63-test suite passed with zero skips**, including optional ZXing and all 15 MariaDB exporter cases, against an isolated socket-only MariaDB 11.8.6 instance. This covers import, rollback, idempotency, printer modules, product lookup, and Y486AP setup; the temporary server was stopped afterward. The source contains 2 catalog items and 2 assets; its launcher defaults to loopback. A read-only live SoR check found source locations `Bin #5` and `Nook`, versus duplicate destination labels `Painted Fern` at IDs 9103 and 9104. No scanner inventory rows were written; physical location mapping remains unresolved. The authenticated browser E2E passed against a disposable SoR and covers export fixture import, duplicate-name location choice by SoR row ID, preview without writes, apply/readback, RackWire profile creation, CSRF-protected barcode lookup, plan persistence, and stubbed bounded-scan admission. Separate native process/HTTP acceptance passed all 13 checks for loopback scan, C control adoption, monitoring targets, and SoR host/address mirror. The import UI exposes optional category-to-hardware-type overrides alongside location and model matching; it states the case-insensitive exact-match default, the `other` fallback, and that overrides affect only new models. The API tests exercise anonymous/viewer rejection, CSRF, required preview fingerprint, category mapping, stale-plan conflict, idempotent replay, and a 5,000-asset preview under a 128 MiB PHP memory limit without writing inventory. The dashboard/API cap is 4 MiB and the paired scanner-import manual is updated. The app now has three local candidate commits on `integrate/solarinet-20261021`; no GitHub or Forgejo remote exists. Physical location mapping, deployed/live import, independent cross-lab review, and subproject release remain open.
- **OBSERVED** Five DNS generation tests, fifteen Khadas DNS/verifier tests, and sixteen isolated BIND release/cutover tests pass. The candidate validates config, compares its generated zone set with Xenon's include-expanded primary-zone inventory, switches one pointer, checks serving, and rolls back with forward-only serials. Legacy cutover recovery is covered. Live chlorine serves 109 zones; its refresh verifies SOA serials for all 109 plus Radium's forward zone, and canonical BIND hashes prove full content and SOA parity across all 109. Recursion is refused and step-ca remains active. The live SoR applier still uses its legacy deployment path; its updated bare-zone generator passes isolated tests but has not been exercised by a live event. Client resolver failover and cross-lab review remain open.
- **VERIFIED LOCALLY 2026-10-21** RackWire is initialized at the integration gitlink's pinned revision `5f29f59`. Its standalone backend suite passes 45 tests; the SolariNet route contract passes 58 checks and the embedded UI bridge passes 51. The integrated browser E2E passes with a disposable SoR and cached Chromium, covering scanner import, profile save, barcode-to-SoR lookup, plan persistence, and stubbed scan admission. The standalone headless smoke covers offline boot, dialogs, local plan persistence, navigation, and narrow layout. RackWire user and developer manuals are single-file HTML with inline styles/SVG and matching US Letter PDFs (7 and 6 pages). Live subnet scan/adoption, deployment, and live SoR commit remain unverified.
- **VERIFIED LOCALLY 2026-10-21** Headless Chrome 156 + Playwright 1.64.1 exercised the standalone `file://` planner with no HTTP requests. The smoke covers accessible Gear library, search, scan, and help dialogs; modal focus trapping, Escape close, and focus restoration; local gear placement and persistence across reload; Map/Data navigation; and a 390 px toolbar with no horizontal page overflow. The test passes locally and is wired into CI; hosted execution awaits the public RackWire GitHub mirror. This does not verify the authenticated SolariNet iframe, live telemetry, or SoR commit/rollback.
- **VERIFIED LOCALLY · 2026-10-21** The SolariNet Interface Guide Rev 2 is now paired with an 18-page US Letter PDF; the HTML remains a single file with inline CSS, JavaScript, and SVG and no linked assets. The contrast matrix has static markup for print/no-JavaScript readers and is recalculated by its runtime. Chrome's own manual checks report no page overflow, density flags, or type-floor flags. The E03/E18 copy and E18 table spacing were tightened to fit; visual checks of pages 1, 3, 6, 17, and 18 show the logo, computed ratios, tables, and footer without overlap. A read-only live Chrome session showed the dashboard SSO screen and RackWire planner; no authentication or mutation was attempted.
- **OBSERVED** The dashboard's SoR host mirror now wraps entity/IP create and soft-delete in one transaction, serializes same-IP operations with a MariaDB advisory lock, and remains idempotent on retries. A scratch-schema integration test injects failures in each multi-step path and races concurrent upserts; wider F-22 coverage remains open.
- **DOCUMENTATION ARTIFACT AUDIT · 2026-10-21** Current named operator/developer manuals and operational guides in core docs, the panel app, MCP, RackWire, and scanner-ingest have paired HTML and PDF files. Every checked PDF is US Letter; the HTML files have no linked stylesheet, script, or image assets. The scanner checkout also has two duplicate architecture/interface HTML copies without local PDFs; their canonical SolariNet docs have paired PDFs. This audit checks packaging, not whether every module README is current or whether deployed behavior matches the manuals.

- **VERIFIED LIVE READ-ONLY COMPARISON · 2026-10-21** The UniFi topology collector's `--compare-sor` path ran against the configured UniFi API and primary SoR. It returned one Default network at `10.0.0.0/8`, no overlap within the source snapshot, and 16 overlapping existing SoR subnets. The comparison marked the network `review_required` for the broad-prefix overlap, no exact SoR subnet, missing UniFi external reference, and missing VLAN tag. It also read SoR DHCP ranges, found no overlapping existing range, and performed no writes. Fifteen UniFi unit tests pass; the read-only transaction contract was tested against a disposable SoR schema. **UNVERIFIED** The configured UniFi connection did not provide a CA file, so TLS certificate authenticity was not checked during this observation. The CIDR conflict remains an import gate; the live report is not approval to alter either topology.

3. Priority sequence and executable gates

1. **Contain and authenticate.** Complete F-02/F-03/F-13/F-14; test viewer, operator, origin, token, and offline states in a real browser. Cut over the PHP/host queue only after the uid split and a rehearsed rollback. A green route test alone is insufficient.
2. **Make state durable.** Close F-08 event delivery, F-10 convergence truth, F-12 migration ledger, and F-17 validated DNS apply. Isolated tests now cover consumer connection loss, RabbitMQ application and container stop/start with a live consumer, and SIGKILL after target apply but before acknowledgement; still exercise host restart, automatic systemd consumer restart, multi-target recovery, and the deployed chain. No event may be acknowledged before its required target state is durable.
3. **Prove identity and lease behavior.** Bind certificates to node IDs and roles (F-01), move long scans off the lease loop (F-05; see the scan worker contract), then independently review and exercise the F-09 lease correction. On 2026-10-21 a fresh native server build passes CTest 37/37, including lease loss between bounded batches and cooperative cancellation of a fake long-running enrichment tool before result writes; the live MariaDB suite passes 14/14, and a disposable worker process test verifies SIGTERM cancellation, lease-loss fencing, standby refusal, orphan recovery, and successful successor scanning with the ASan/UBSan binary. The lease suite passes eight synchronized two-claimant expiry races using separate database handles, and a process test confirms a second real server stays standby, then acquires epoch 2 and rebinds all listeners after the first exits. Migration 028 and the generated schema snapshot match. Dashboard route/UI, release-package, and installer/rollback tests pass. The worker remains undeployed. Systemd-managed stop timing, deployed enrichment binaries, deployed-pair failover under host/network timing, bind-failure behavior, standby mutation gating, cross-lab review, production migration, and rollback rehearsal remain open.
4. **Integrate deliberately.** Advance RackWire's pin, preserve its standalone SQLite/discovery boundary, and test the SolariNet iframe/API/SoR commit boundary. Add shared maturity, provenance, freshness, and entity navigation as specified by I0-I4 in the August workflow.
5. **Publish only after evidence.** Produce self-contained HTML user/developer manuals and parallel Letter PDFs for each app; review release artifacts and run restore/rollback drills. Merge to Forgejo main, mirror GitHub, deploy staged artifacts, verify actual services, then tag matching releases.

> **Working contract:** `docs/SolariNet_Stabilization_Integration_Workflow_2026-08-09.md` defines F-01 through F-31 and I0-I5, including scanner-ingest intake. RackWire's current `docs/CONTRACT-RW.md` supersedes the two retired signal-token assertions. The stale root README phase table and netdb YAML-authority text are corrected in this candidate, pending merge.

4. Unverified and blocked

- **UNVERIFIED** Automatic UniFi topology import remains unimplemented. The live read-only comparison on 2026-10-21 found one source `10.0.0.0/8` network overlapping 16 SoR subnets; preserve the existing SoR rows until topology authority and mapping are resolved. The configured UniFi TLS certificate was not validated during that observation.

- **VERIFIED · 2026-10-21** Rebuilt the scan worker and C control bridge, then replayed the browser flow from authenticated RackWire through its **Network discovery** handoff and Discovery UI against a fresh temporary MariaDB with all 34 monitoring migrations. The worker scanned `127.0.0.1/32` against a temporary TCP listener; the UI observed admission, queue execution, status polling, navigation recovery, and a reviewable candidate. CI runs the loopback path with pinned Playwright/Chromium. A separate native-worker scan of directly connected gateway `10.0.0.1/32` persisted one candidate in scratch. The default release build completed a real scan of directly connected `10.0.0.0/24`: **254/254** addresses processed, **13** candidates persisted, and terminal status in **10.32 seconds** against a fresh scratch schema. The opt-in RackWire-to-worker browser flow then completed the same /24 through the PHP API and C queue, displayed a local-interface candidate after navigation recovery, and observed **14** candidates. The progress message now distinguishes host scanning from candidate finalization and reports candidates saved; the browser acceptance verifies the transition. Both temporary workers, databases, and accounts were removed; no live monitoring or SoR data was written. Optional enrichment now limits Nmap to ports found by the native sweep, uses light service probes, and limits OS detection. With Nmap, SNMP, and Avahi enabled, the worker completed a directly connected `10.0.0.0/24` in **136.34 seconds**: 254/254 hosts and 12 candidates in a disposable monitoring schema. The temporary DB, account, worker, config, and lock were removed. **UNVERIFIED** Other VLANs, deployed worker behavior, and browser polling during enrichment remain open. Discovery adoption mirrors a SoR host entity/address, not a physical hardware unit.
- **UNVERIFIED** Authenticated browser writes against deployed services, queue cutover on xenon, RackWire iframe actions against live SolariNet, SoR propagation through all production render targets, migration adoption on current databases, and Xenon's monitoring database backup/restore. Isolated browser acceptance covers both scanner → SoR → RackWire profile → barcode → plan-save and RackWire → native scan worker → Discovery candidate flows.
- **VERIFIED LOCALLY** The current cross-build matrix configured and built Linux arm64, Linux arm32, and Windows x86_64 with the checked-in toolchains. Linux arm64 and arm32 test executables passed under QEMU; Windows was build-only, matching CI. This does not verify Windows runtime behavior or cross-build optional I/O dependencies.
- **UNVERIFIED** Fleet-scale monitoring, three-monitor failover, public ACME challenge success, and documentation freshness across all modules and parity with deployed behavior.
- **OBSERVED ON XENON · 21 OCTOBER 2026** Five active SoR sync/notification units have restart-limit settings in `[Service]`, so systemd ignores the configured 60-second interval and applies its 10-second default. Candidate unit files now place the settings in `[Unit]`; a CI parser checks all 43 repository service files. Live unit replacement and effective-limit verification remain part of staged deployment.
- **OBSERVED** Claude CLI cross-lab review could not run because its OAuth session had expired. Independent review is still an open release gate for the new security and integration work.